CS-T35-DATA STRUCTURES
2MARKS
UNIT-1

1. Define Data Structures.
2. Define an Abstract Data Type (ADT).
3. What are the objectives of studying data structures?
4. What are the components of space complexity?
5. Define time complexity.
6. What are the different asymptotic notations?

UNIT-2
LINKED LIST

1. Define Linked Lists
2. State the different types of linked lists
3. List the basic operations carried out in a linked list
4. List out the advantages of using a linked list
5. List out the disadvantages of using a linked list
6. List out the applications of a linked list
7. State the difference between arrays and linked lists

STACKS

1. Define a stack .
2. List out the basic operations that can be performed on a stack.
3. State the different ways of representing expressions
4. State the rules to be followed during infix to postfix conversions .
5. State the difference between stacks and linked lists .
6. Mention the advantages of representing stacks using linked lists than arrays.

QUEUE

1. Define a queue
2 Define a priority queue
3. State the difference between queues and linked lists
4. Define a Deque
5. What are the types of queues?
6. List the applications of stacks .
7. List the applications of queues.

UNIT-4
TREE STRUCTURES

1. Define tree.
2. Define Root .

3. Define Degree of the node
4. Define Leaves.
5. Define internal nodes.
6. Define Parent node.
7. Define Depth and height of a node
8. Define Level of the tree
9. Define Forest.
10. Define Path.
11. Define a binary tree.
12. Define a full binary tree.
13. Define a complete binary tree.
14. State the properties of a binary tree.
15. What is meant by binary tree traversal?
16. What are the different binary tree traversal techniques and what are the task performed during traversals.
17. State the merits of linear representation of binary trees.
18. State the demerit of linear representation of binary trees.
19. State the merit of linked representation of binary trees.
20. State the demerits of linked representation of binary trees.
21. Define a binary search tree.

BALANCED TREES

1. Define AVL Tree.
2. What do you mean by balanced trees?
3. What are the categories of AVL rotations?
4. What do you mean by balance factor of a node in AVL tree?
5. Define Heap.
6. Define B-tree of order M.
7.What are the applications of B-tree.
8. What are the properties of binary heap?
9. What do you mean by structure property in a heap?
10. What do you mean by heap order property?
11. What are the applications of priority queues?

UNIT-IV
GRAPH TERMINOLOGIES

1. Define Graphs.
2. Define Adjacent nodes.
3. Define Directed graph.
4. Define Undirected graph
5. Define Loop.
6. Simple graph.
7. Define Weighted graph.
8. Define Outdegree of a graph.
9. Define Indegree of a graph.

10.Define path in a graph
11.Define Simple path.
12.Define cycle or a circuit.
13.Define Acyclic graph.
14.Define Strongly connected in a graph.
15.Define Weakly connected.
16.Name the different ways of representing a graph?
17. What is an undirected acyclic graph?
18. What are the two traversal strategies used in traversing a graph?
19. What is a minimum spanning tree?
20. Name two algorithms two find minimum spanning tree
21. Define graph traversals.
22. List the two important key points of depth first search.
23. What do you mean by breadth first search (BFS)?
24. Differentiate BFS and DFS.
25. What is meant by topological sort?

UNIT-V

1. Define sequential files.
2. What are the Methods used for transaction logging?
3. What do you mean by indexed sequential files?
4. Define a record.
5. What is Symbol table?
6. What is Hashing function?
7. What is meant by Folding Method?
8. Define MTTF.
9. Define hash tables.
10. Define Jagged table.

11 MARKS

UNIT-1

1 . A)Describe the criteria to be satisfied by an algorithm.(5)(Nov 2014,2010)

B) Illustrate the algorithmic notations.(5)(April 2012)

2. Explain 1d-array,2d-array.(april 2013,april 2012)
3. Explain Linear and binary search in detail.
4. Explain (i) Quick Sort (ii) Merge Sort (iii) Shell sort with Example.
5. Explain Heap sort.

UNIT-II

1. Write the equivalent postfix expression for the following expressions using expression evaluation algorithm) (A-B)/(D+E)*F) ii)((A+B)/D) (E-F)*G (Dec2014)
2. Compare and Contrast Stack Vs Queue.
3. Implement a queue by singly linled list l and perform the operation enqueue and dequeue. (May 2012).
4. What is doubly linked list? Explain the insertion and deletion operation of double linked list with algorithm & swapping of two nodes. (Nov 2012, April 2013, Nov 2013,Dec 2014).
5. Explain dynamic storage management.
6. Discuss stack and its operations.
7. Explain Queue and its operations.
8. Explain (i) Dequeue (ii) Circular Queue.

UNIT-III

1. Explain the traversals binary tree traversal with examples.(April 2014,Dec 2014,Nov 2015)
2. Explain the B+ tree indexing and trie tree indexing.(Nov 2015,April 2014,May 2012).
3. Discuss AVL Trees in detail.
4. Explain the Dijikstra’s algorithm to find the shortest path. (Nov 2010,April 2012,Nov 2013)
5. Discuss about tree and its terminologies.

UNIT-IV

1. Discuss about Graph traversal.(April 2013,Nov 2013)
2. Describe the topological sort with an example.(Nov 2014,2010,April 2014)
3. Explain Sets in detail.
4. Explain Graphs and its terminologies.

UNIT-V

1. Explain the collision resolution techniques in hashing. (Nov 2014,April 2015)
2. Distinguish between sequential and direct files. List out the properties & of an indexed sequential file and explain.(Nov 2014,Nov 2010)
3. Explain the following tables, its use and representation in detail(i) Jagged table,(ii) Symbol table(iii) Hash table(April 2014)